

SUMANASRI SURISETTI

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EDUCATION

Florida Institute of Technology, Melbourne, FL

May 2028

Bachelor of Science in Astronomy and Astrophysics, Minor in Computational Mathematics

Relevant Coursework: Computational Methods, Partial Differential Equations, Electromagnetic Theory, Parallel Processing

EXPERIENCE

Red Canyon Engineering & Software, Denver, CO

June 2026–Present

cFS Intern

- Extend cFS applications with new command codes, handler functions, and event messages in C, verifying behavior end to end by sending commands through the cFS ground system and confirming event and telemetry output
- Integrate and extend a UT-Assert stub generator that replaces cFE and OSAL dependencies, enabling cFS application logic to be unit-tested in isolation
- Integrate applications into cFS mission builds through CMake target lists and cFE startup scripts, commanding cFE core services (Executive, Software Bus, Event, Time, Table) through the Python ground system to verify runtime behavior on Linux

Research Advisor: Dr. Sofiane Bourouaine, Florida Institute of Technology, Melbourne, FL

March 2026–Present

Space Physics Research Assistant

- First-author analysis accepted to the AAS Solar Physics Division 57th meeting (SPD 57), August 2026
- Convert Parker Solar Probe SWEAP/SPC instrument measurements into visualizations and model-ready datasets, screening for quality flags and instrument artifacts before analysis
- Built a PFSS and OMNI mapping pipeline testing source-surface height sensitivity at 2.0, 2.5, and 3.0 solar radii against near-Earth solar wind observations across five Carrington rotations

Astronomical Research and Engineering Society, Ortega Observatory, Melbourne, FL

September 2023–Present

Research Associate

- Instructed students in observational dataset analysis using computational methods including FITS file processing and spectroscopy, and reviewed student datasets and research documentation for unit, metadata, and method errors

PROJECTS

Spacecraft Attitude and Proximity Operations Simulation

C, Python, SQLite, Makefile

- Built an HWIL-style attitude control simulation separating Python plant dynamics from a C flight software controller with an RTOS-style scheduler running sensor, control, telemetry, and health tasks at 100, 50, 10, and 1 Hz
- Implemented a per-axis Kalman filter with a gyro-bias state, deriving process noise from the modeled disturbance torque with no free tuning, cutting steady-state pointing error RMS from 9.0×10^{-3} to 1.2×10^{-3} rad and torque activity 3x relative to raw sensors
- Extended the simulation to proximity operations with Clohessy-Wiltshire dynamics in the Hill frame, quantifying the 0.68 m/s versus 2.07 m/s delta-v cost of closed-loop glideslope guidance against a two-impulse transfer on a 500 m $\rightarrow$ 50 m V-bar approach
- Implemented onboard abort logic evaluated on the estimated relative state with a 30 m keep-out sphere and closing-speed corridor, aborting a mis-commanded approach at $t = 1474$ s with 39.9 m minimum true range

High-Power Rocket Design and Launch, Panther Rocket Lab, Melbourne, FL

- Designed, built, and launched a high-power rocket under Level 1 certification requirements, laser cutting the wood fins and slotted tube frame parts, 3D printing the nose cone and internal component mount, and epoxy bonding the assembly
- Integrated Raspberry Pi and Arduino GPS modules for onboard position tracking, verifying electronics operation, recovery setup, mechanical fit, and failure points during launch-readiness checks

Astrophysics Analysis Pipelines

Python, NumPy, Astropy, SunPy, scikit-image

- Compared Lomb-Scargle and Box Least Squares period recovery on Kepler photometry and injected-transit data, recovering a 3.72 day injected period to within 1.4 minutes
- Propagated two-body orbits with Euler, semi-implicit Euler, and velocity Verlet across timesteps from 1 s to 300 s measuring energy and angular momentum drift, and built batch lunar image quality analysis computing 17 metrics per image

SKILLS

Languages & Tools: Python, C, C++, SQL, Bash, Linux/Unix, Git, GitHub Actions (CI/CD), CMake, Makefile, CTest, SQLite, Jira

Spacecraft Software: NASA cFS/cFE, ground system commanding, telemetry verification, anomaly detection, unit and integration test

Data & Scientific Computing: NumPy, Pandas, SciPy, Astropy, SunPy, Matplotlib, FITS handling, Monte Carlo methods

Hardware & Prototyping: Arduino, Raspberry Pi, sensor integration, 3D printing, laser cutting, test automation